

A Comparative Study Using Multinomial and Ordered Logit Models

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Abstract

This study focuses on the comparison of ordered and unordered models, namely which are the Ordered Logit Model and the Multinomial Logit Model. This comparison is based on which model is more suitable or best to handle a given qualitative dataset, and a qualitative dataset can either be nominal or ordinal. The dataset to be used in this paper is the APS (Adolescent Placement Study) which is a qualitative dataset. The Wald will be used to determine which model best suits this dataset since it can be applied to both the tests. Throughout the research, it was determined through various procedure which model would be best to model the dataset we have. Those methods included determining the trend of the parameter estimates in each model, seeing if the estimated odds would be explained in the same way and also if the trend of the predicted probabilities is the same in either model. But now it was found that the proportional odds assumption of the ordinal logit model was violated when all the explanatory variables were included in the model. But now from all of these factors considered, it was found that the Multinomial Logit Model was the better model to model this APS dataset due to its flexibility and better interpretation of the results.

Declaration

I, *Siyabonga Mahlangu* declare that this essay, submitted in partial fulfillment of the degree *BSc(Hons) Mathematical Statistics / BCom(Hons) Mathematical Statistics / BCom(Hons) Statistics*, at the University of Pretoria, is my own work and has not been previously submitted at this or any other tertiary institution.

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1 Introduction

Statistical models play a very important role in analysing and making predictions on the factors that impact our daily lifestyles. Within those models, the ordered logit model and multinomial logit model are used quite often in analysing such factors, but are limited only to response variables with three or more categories.

Until now, there still exists a gap in a comparative study of these two models based on their performance on factors that impact our daily lifestyles. Although these models are very capable in their own right in handling the above mentioned data, problem is they use different assumptions and methodologies leading to variations in predictive accuracy and model fit. This is the reason why there needs to be a study to evaluate both their strengths and weaknesses.

In this paper, we would like to better understand each model and address the gap between the models based on a comparative analysis. We will conduct this study by addressing the following questions:

- How does the ordered logit and multinomial logit models differ in terms of their theoretical foundations and assumptions?
- What are the methodological differences between these two models in an application process?
- To what extent do these models accurately predict what they need to predict based on empirical data?
- How do these models perform in terms of goodness-of-fit measurements?

By addressing these questions, we will get invaluable insights into the comparative performance of the ordered logit and multinomial logit models. The findings on this paper will thus contribute to the methodology and allowing the community at large to make informed decisions in the aspect of what model to use.

Now this research boils down to the following question: 'Does the nature of the data (Nominal or Ordinal) affect the choice of model selection?' But for now we are going to dive more in detail about the background of the two models.

2 Literature Review

In 2019, Lai and Chan conducted a study about "Comparison of the linear regression, multinomial logit, and ordered probability models for predicting the distribution of thermal sensation" [1]. In this study, they wanted to know which model between the three yielded better results on the "distribution of thermal sensation". To achieve this, they used two datasets of which one was from an indoor thermal comfort survey conducted in Pakistan and the other an outdoor conducted in Tianjin in the country of China.

The datasets were split into two parts which are the training(for model development) and testing (for predictions) parts. To assess the accuracy of the models, they used the "percentage of correct predictions". The results of this paper showed that the "multinomial logit model was slightly more accurate than the ordered probit model". Moreover, between the three models being studied, the chi square test also proved that the ordered and the multinomial logit model proved to be more accurate than the other model. Also, the linear regression model correctly estimated 20 to 40% of predictions whereas the other two models predictions were about 50%. Thus showing us that the type of dataset does impact the choice of model selection.

The study by Lai and Chan can be supported by Ye and Lord on a study they did in 2014 about "Comparing three commonly used crash severity models on sample size requirements: Multinomial logit, ordered probit and mixed logit models" [2]. To accomplish the objectives ("examine the effects of the sample size on the models mentioned above") of the mentioned paper, a simulated and the observed crash datasets were used, in which the observed dataset had 27 independent variables. The study shows that small sample size does affect model development on all three models, thus having a dataset with a sufficient sample size is crucial. The results of the study yielded the following: when estimating the multinomial model and testing at 5% level of significance, out of 27 independent variables only 10 were retained. And When using the probit logit model, out of the 27 variables, only 18 were retained at 5% level of significance. Making the multinomial logit model more preferred than the ordered logit model, this is because of it has more interpretative power. In terms of the goodness of fit test, the ordered logit model proved to be superior as it had more included explanatory variables compared to others, making the above models better than the other one.

According to Akkus and Ozkoc, on a study about A "Comparison of the Models Over the Data on the Interest Level in Politics in Turkey and Countries that are Members of the European Union: Multinomial or Ordered Logit Model?"[3]. Now they first made a strict assumption Independence of Irrelevant Alternative (IIA) which for example says, if A is chosen over B, then introducing another option must not result in B chosen over A. The significance of the estimates were tested at 5% level of significance. This assumption led them to conclude that they cannot apply the multinomial logit model to the data they had, this is due to the fact that the IIA assumption was the most important and it was not provided in the model. And the parallel slope assumption was important in the ordered logit model which the results yielded that the ordered logit model was more fit to be applied in the dataset used, and also, they showed that at 5% level of significance the ordered logit model was significant by using a chi-square distribution having 18 degrees of freedom. Moreover, it was shown that the effects of the explanatory variables on the response variable were significant on the ordered logit model than on the multinomial logit model, making the Ordered Logit Model better.

Not forgetting about the study about Comparison of Vehicle-Ownership Models done by Potoglou and Susilo in 2008 [4]. This study used three observed datasets to achieve the objective of "Which model would better reflect households's car ownership choices?". The discrete choice models (the ordered logit model is one of those) were evaluated using the measures of data fit which are the AIC, BIC and HQIC. The study resulted in all models performing very well since they yielded high values of the goodness-of-fit. But based on the adjusted likelihood ratio values, the multinomial logit model was determined to be the best from the rest as it showed to have the highest value of the index and this was also supported by the measures of data fit.

Up to this point, it shows that there is a significant lack of studies only between the Multinomial and the Ordered Logit Models, and in addition, it also proves that both the models are very useful in their own right. In this paper, the most accurate model will be determined by an accepted measurement of performance.

3 Binary Logit Model

Before get to more information about our first model, the Multinomial Logit Model, we first have to understand the Binary Logit Model. This model was founded by a well known statistician David Cox [5]. He founded this model in the year 1958 for the purpose of trying to find a way to model binary outcome variables such as yes/no, success/failure [5]. For us to understand the derivations to be done in the Multinomial Logit Model, we first have to understand the derivations in the Binary Logit Model.

Let Y be a binary outcome variable, where:

$$Y = \begin{cases} 0 & \text{if NO} \\ 1 & \text{if YES} \end{cases}$$

Given the binary outcome variables, the logit model is:

$$\log \left(\frac{P(Y = 1)}{P(Y = 0)} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p \quad (1)$$

The Binary Logit Model uses the log-odds to model the relationship between the explanatory and the response categorical variable where the odds can be expressed as:

$$\begin{aligned}
\text{Odds} &= \frac{P}{1 - P} \\
&= \frac{\text{number of events}}{\text{number of non-events}} \\
&= \frac{\text{probability of an event}}{\text{probability of a non-event}} \\
&= \frac{P(Y = 1)}{P(Y = 0)}
\end{aligned} \tag{2}$$

with P being the probability of an event occurring.

The Odds equation can be manipulated to make the probability of an event occurring the subject of a formula as follows:

From (2) it follows that:

$$P(Y = 1) = \text{Odds} \cdot P(Y = 0) \tag{3}$$

Do not forget that the sum of probabilities should be equal to one, given that knowledge, we can again manipulate all the information we know thus far to make the probability of an event not occurring the subject of a formula as follows:

$$P(Y = 0) + P(Y = 1) = 1 \tag{4}$$

By substituting (3) into (4), it follows that:

$$P(Y = 0) + \text{Odds} \cdot P(Y = 0) = 1 \tag{5}$$

taking out the common factor, it leads to:

$$P(Y = 0)(1 + \text{Odds}) = 1 \tag{6}$$

$$P(Y = 0) = \frac{1}{1 + \text{Odds}} \tag{7}$$

And by substituting (7) into (1), we get:

$$P(Y = 1) = \frac{\text{Odds}}{1 + \text{Odds}} \tag{8}$$

Above is another way to calculate the probability of an event occurring.

Mind you, we first defined our logit model in (1) as well as our Odds in (2), by fitting the model, we can get our estimated log-odds equation. The estimated log-odds equation for a logistic regression model

is given by:

$$\log(\text{Odds}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p \quad (9)$$

Now by exponentiating/taking the anti-log of the fitted model, we can also determine our estimated odds which can be expressed as follows:

$$\text{Odds} = \exp(\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p) \quad (10)$$

The estimated probabilities can now be determined since we have the estimated odds by means of the equations in (7) and (8).

4 Multinomial Logit Model

The multinomial logit model was created by Daniel McFadden an American economist, well known for his work in the fields of econometrics and discrete choice modeling [6]. He developed this model in the 1970s as part of his wider research on discrete choice analysis [7]. This model is important in the fields such as social sciences, marketing or economics and many more where modeling choices of multiple alternatives is of significance [8]. The model itself goes by a different variety of names such as sometimes being called polytomous or polychotomous [9]. But these names are industry based.

The multinomial logit (log of the odds) model is a model used to predict the probabilities of the different outcomes of the j^{th} categorical response variable given a set of independent variables which may be binary or real valued, this can be used to "determine the percentage of variance in the response variable explained by the explanatory variables" [8].

Similar to other models, the multinomial logit model also has a known number of assumptions which it must follow. A few of those assumptions are [10]:

- No Perfect Multicollinearity - This is because perfect correlation between explanatory variables would make it impossible to determine the unique impact of each explanatory variable on the response variable.
- Independent Observations [11] - The choice made by one individual should not be influenced by another individual.
- Linearity - The logit of the probabilities is a linear combination of the predictor variables.

Even though the model has such assumptions, in addition, it has a few known advantages and disadvantages.

The advantages are [12]:

- "Flexibility: Multinomial logistic regression can be used for a wide range of dependent variables with multiple categories. It can handle both nominal and ordinal categorical data."
- "Interpretability: The model provides coefficients for each independent variable that can be used to interpret the effect of the independent variables on the dependent variable."
- "Efficient Computation: The optimization algorithms used for multinomial logistic regression are efficient and can handle large datasets."

With the disadvantages being [13]:

- Sample Size Requirements [14] - Multinomial logit regression often requires a large sample size, especially when there are numerous predictor variables or response categories. Having too few cases in any category can lead to unreliable parameter estimates.
- Parameter Complexity - This model involves many parameters, many of which may not be of primary interest. While the loglinear model explains the joint distribution of all variables, the logistic model focuses solely on the conditional distribution of the response given the predictors.
- Interpretation - The loglinear model can be more challenging to interpret compared to the logistic model. In the loglinear approach, the influence of a predictor X on the response Y is represented by the association between X and Y.

The multinomial logit model is designed for data that does not have any inherent ordering in it [15]. For example, the outcome variable can be favorite color with categories 'blue', 'red' or 'green', it can be noted that there is no natural order in these colors.

The model uses a multinomial probability distribution for the outcome variable where the model is a set of binomial/binary logit models. This is because of the situations where the response variable has more than two categories. In addition, the model uses Maximum Likelihood Estimation to estimate the parameter estimates [15] in which the parameters to be estimated in this paper will be estimated using SAS.

Moreover, the model uses the log-odds where $\text{odds} = \frac{p_i}{1-p_i}$ where p_i is the probability of an event occurring. The use of log-odds provides a way to model the relationship between the explanatory and multinomial response variable allowing for the estimation of the probabilities in a manner that adheres to the constraints of the probabilities which are:

- The sum $\sum_{i=1}^n p_i = 1$.
- value of p_i satisfies $0 \leq p_i \leq 1$.

Additionally, this transformation allows the relationship to be modelled linearly, this in turn simplifies the estimation process making it possible to use linear regression techniques on the transformed variable.

The following are the general model specifications of the multinomial logit model. Keep in mind that in the next section, the derivations used to arrive to these general specifications of the model are explained more in detail. Moreover, in this paper, our reference category will be 0.

The multinomial logit model equation is:

$$\log \left(\frac{P(Y = j)}{P(Y = 0)} \right) = \beta_{j0} + \beta_{j1}x_1 + \beta_{j2}x_2 + \dots + \beta_{jp}x_p \quad \text{for } j = 0, 1, 2, \dots, J \quad (11)$$

Where $P(Y = j)$ is the probability of each outcome for each observation besides observation $j = 0$, and $P(Y = 0)$ is the probability of the reference category.

The regression coefficients in the equation above represent the effects on the log-odds of inclusion in category j against the reference category 0

The probability for category j in the multinomial logit model is:

$$P(Y = j) = \frac{\exp(\beta_{j0} + \beta_{j1}x_1 + \dots + \beta_{jp}x_p)}{1 + \sum_{j=1}^J \exp(\beta_{j0} + \beta_{j1}x_1 + \dots + \beta_{jp}x_p)} \quad \text{for } j > 0 \quad (12)$$

and since $\sum_{i=1}^J p_i = 1$ it follows that the probability of the reference category can be expressed as:

$$P(Y = 0) = \frac{1}{1 + \sum_{j=1}^J \exp(\beta_{j0} + \beta_{j1}x_1 + \dots + \beta_{jp}x_p)} \quad (13)$$

where the reference category can be used to avoid redundant parameter estimates. The total number of parameters to be estimated in a multinomial logit model are $2(p + 1)$.

4.1 THREE OUTCOME CATEGORIES

In this part of the research, we are going to derive the model specifications of the Multinomial Logit Model using three categories as possible outcomes. This will in detail give us the reason how we defined the general model specifications above. Keep in mind that a multinomial logit model is limited to more than two categories.

Let Y be an outcome variable with 3 categories, where:

$$Y = \begin{cases} 0 & \text{if NO} \\ 1 & \text{if MAYBE} \\ 2 & \text{if YES} \end{cases}$$

In this case, two Multinomial Logit Models will be fitted and solved simultaneously to get the estimated

model. If the response variable had four outcome categories, three multinomial logit models would be fitted and solved simultaneously to get the estimated model. Moreover, our reference/baseline category will be $Y = 0$ in any case.

The first logit model to be fitted is defined as:

$$\log \left(\frac{P(Y = 1)}{P(Y = 0)} \right) = \beta_{10} + \beta_{11}x_1 + \beta_{12}x_2 + \dots + \beta_{1p}x_p \quad (14)$$

The second logit model to be fitted is defined as:

$$\log \left(\frac{P(Y = 2)}{P(Y = 0)} \right) = \beta_{20} + \beta_{21}x_1 + \beta_{22}x_2 + \dots + \beta_{2p}x_p \quad (15)$$

4.2 ODDS AND PROBABILITY EXPRESSIONS

Since the Odds has been defined, it is safe to say that there will be two Odds expressions we will be dealing with.

The Odds for category 1 can be expressed as follows:

$$\text{Odds}_1 = \frac{P(Y = 1)}{P(Y = 0)} \quad (16)$$

And the Odds for category 2 can be expressed as follows:

$$\text{Odds}_2 = \frac{P(Y = 2)}{P(Y = 0)} \quad (17)$$

Now the probabilities of category 1 and 2 will be as follows:

$$P(Y = 1) = \text{Odds}_1 \cdot P(Y = 0) \quad (18)$$

And

$$P(Y = 2) = \text{Odds}_2 \cdot P(Y = 0) \quad (19)$$

But not forgetting that;

$$P(Y = 0) + P(Y = 1) + P(Y = 2) = 1 \quad (20)$$

By substituting (18) and (19) into (20), we get the following expression:

$$P(Y = 0) + \text{Odds}_1 \cdot P(Y = 0) + \text{Odds}_2 \cdot P(Y = 0) = 1 \quad (21)$$

And by taking out the common factor as well as simplifying the expression, we get:

$$P(Y = 0)(1 + \text{Odds}_1 + \text{Odds}_2) = 1$$

$$\begin{aligned} P(Y = 0) &= \frac{1}{1 + \text{Odds}_1 + \text{Odds}_2} \\ &= \frac{1}{1 + \exp(\beta_{10} + \beta_{11}x_1 + \dots + \beta_{1p}x_p) + \exp(\beta_{20} + \beta_{21}x_1 + \dots + \beta_{2p}x_p)} \end{aligned}$$

Therefore, the probabilities for categories 1 and 2 can also be calculated in the following manner:

By substituting $P(Y = 0) = \frac{1}{1 + \text{Odds}_1 + \text{Odds}_2}$ into (18) and (19).

$$P(Y = 1) = \frac{\exp(\beta_{10} + \beta_{11}x_1 + \dots + \beta_{1p}x_p)}{1 + \exp(\beta_{10} + \beta_{11}x_1 + \dots + \beta_{1p}x_p) + \exp(\beta_{20} + \beta_{21}x_1 + \dots + \beta_{2p}x_p)} \quad (22)$$

and

$$P(Y = 2) = \frac{\exp(\beta_{20} + \beta_{21}x_1 + \dots + \beta_{2p}x_p)}{1 + \exp(\beta_{10} + \beta_{11}x_1 + \dots + \beta_{1p}x_p) + \exp(\beta_{20} + \beta_{21}x_1 + \dots + \beta_{2p}x_p)} \quad (23)$$

Taking from the expressions derived from the Binary Logit Model, the fitted Multinomial Logit Models will be expressed as follows:

$$\log(\hat{\text{Odds}}_1) = \hat{\beta}_{10} + \hat{\beta}_{11}x_1 + \hat{\beta}_{12}x_2 + \dots + \hat{\beta}_{1p}x_p \quad (24)$$

and

$$\log(\hat{\text{Odds}}_2) = \hat{\beta}_{20} + \hat{\beta}_{21}x_1 + \hat{\beta}_{22}x_2 + \dots + \hat{\beta}_{2p}x_p \quad (25)$$

Not forgetting that we can get our estimated odds by taking the anti-log/exponentiating the log-odds equation. This will thus lead to the following expressions:

$$\hat{\text{Odds}}_1 = \exp(\hat{\beta}_{10} + \hat{\beta}_{11}x_1 + \hat{\beta}_{12}x_2 + \dots + \hat{\beta}_{1p}x_p) \quad (26)$$

and

$$\hat{\text{Odds}}_2 = \exp(\hat{\beta}_{20} + \hat{\beta}_{21}x_1 + \hat{\beta}_{22}x_2 + \dots + \hat{\beta}_{2p}x_p) \quad (27)$$

4.3 ODDS RATIO

With all that has been said about the multinomial logit model, we can also determine the odds ratio which is a statistic used to compare the odds of an event happening in one group to the odds of it happening in another group [16]. Also, the Odds Ratio can be used to compare certain levels of the explanatory variable, that way will be shown later in the section [16]. This ratio is commonly used in case control trials or in any research designs where there is a significant association between exposure and outcome [16].

The general formula of the Odds Ratio is:

$$\begin{aligned}
 \text{OR} &= \frac{\text{Odds}_1}{\text{Odds}_2} \\
 &= \frac{P(Y=1)}{P(Y=0)} \div \frac{P(Y=2)}{P(Y=0)} \\
 &= \frac{P(Y=1)}{P(Y=2)} \\
 &= \frac{\exp(\beta_{10} + \beta_{11}x_1 + \dots + \beta_{1p}x_p)}{\exp(\beta_{20} + \beta_{21}x_1 + \dots + \beta_{2p}x_p)}
 \end{aligned} \tag{28}$$

Thus making the estimated odds ratio:

$$\hat{\text{OR}} = \frac{\exp(\hat{\beta}_{10} + \hat{\beta}_{11}x_1 + \dots + \hat{\beta}_{1p}x_p)}{\exp(\hat{\beta}_{20} + \hat{\beta}_{21}x_1 + \dots + \hat{\beta}_{2p}x_p)} \tag{1}$$

The general interpretation of the Odds Ratio is as follows:

- $\text{OR} = 1$: The odds of the event are the same in both groups.
- $\text{OR} > 1$: The odds of the event are higher in the exposed group compared to the unexposed group.
- $\text{OR} < 1$: The odds of the event are lower in the exposed group compared to the unexposed group.

Now below we are going to do a simple example of how the Odds Ratio works and how to reach conclusions based on the calculations we have done. Consider the table below of the association between smoking and lung cancer:

Table 1: Example of Odds Ratio Table

	Lung cancer	No Lung cancer
Smokers	50	150
Non-Smokers	10	190

In order for us to determine the Odds Ratio, we firstly have to calculate the odds for the two groups.

The Odds for the group 1 is:

$$\text{Odds}_1 = \frac{50}{150} = \frac{1}{3}$$

The Odds for group 2 is:

$$\text{Odds}_2 = \frac{10}{190} = \frac{1}{19}$$

With having calculated the Odds for both groups, we can now calculate our Odds Ratio.

$$\text{OR} = \frac{\text{Odds}_1}{\text{Odds}_2} = 6.33$$

This means that the odds of developing lung cancer are approximately 6.33 times higher for smokers than for non-smokers.

5 Ordinal Logit Model

The ordinal logit model was created by well known distinguished individual for the puporse of extending the applicability of the logistic regression to situations where the outcome variable is ordinal. The individual was a Peter McCullagh who created the model in the 1980s. Peter McCullagh made his work well known in statistics, more specifically in the field of Generalized Linear Models (GLM) and the analysis of categorical data.

According to Hosmer and Lemeshow, there are three different Ordinal Logistic Regression Models in which researchers have to choose from, where most do not even know which one to pick when dealing with a specific data. The three models are [31]:

- The cumulative logit model
- The adjacent categories model
- The continuation ratio model

But for the purposes of this research, we will focus on the cumulative logit model, which is also known as ordered logit or ordinal logit model [27].

This model is usually preferred for because of two superior reasons which are [27]:

- "They are simpler, and therefore easier to interpret."
- "Hypothesis tests are more powerful."

Just like the Multinomial Logit Model, the ordered Logit Model is also important in the fields of economics, social sciences, marketing and many more. This model sometimes goes by the name of proportional odds model. As with any other model, there are advantages and disadvantages with this one as well.

Its advantages are:

- Agile - It can handle both categorical and continuous predictor variables.
- Efficient - Since the model uses the logistic function, the model is computationally efficient and can be estimated using any desired software.
- Interpretable - provide highly interpretable coefficients that explain the relationship between your features and your outcome variable.

Its disadvantages are:

- Proportional Odds Assumption - It makes a fairly strong assumption that is not necessarily valid in all cases.
- Limited to Ordinal Outcomes - The model is specifically designed for ordinal outcomes and is not suitable for nominal outcomes (unordered categories) or continuous outcomes.
- Sensitivity to Outliers - The model can be sensitive to outliers and influential data points, which can distort the results and lead to misleading conclusions.

Even though the model has such advantages and disadvantages, according to Richard Williams, there is a trend in which researchers apparently do not follow the assumptions of this model [28]. Not following the assumptions of the model might lead to researchers obtaining biased or inconsistent estimates. And just to name a few, the model assumptions are as follows:

- The dependent variable must be ordinal, its categories have a meaningful order.
- The independent variables should not be perfectly correlated with each other.
- The observations should be independent of each other

Keep in mind that if the above assumptions, and even the ones not stated above are met, "then all of the corresponding coefficients (except the intercepts) should be the same across the different logistic regressions, other than differences caused by sampling variability and the odds ratios will stay the same regardless of which of the collapsed logistic regressions is estimated (hence the term proportional odds assumption is also often used)" [28].

The model can handle categories that have inherent ordering in it such as educational attainment [29]. The order of educational attainment goes from No formal education, Primary school, High school diploma, Associate degree, Bachelors degree, Masters degree, and finally to Doctorate. It can be noted that the levels naturally progress from lower to higher education. But take note that if the response variable only has two categories, ordered or not, the model will reduce to a binary logit model [27].

Take note that the model uses the cumulative probabilities of ordinal outcomes, the model specifically models the probability that the outcome variable Y is less than or equal to a certain category j , where j is one of the possible ordered categories of the dependent variable. Moreover, the model assumes that the log-odds of being in category j or lower is a linear function of the predictors variables [29]. The parameter estimates are also estimated using the Maximum Likelihood Estimation just as in the multinomial logit model [30].

The following are the general model specifications of the ordered logit model. A more detailed explanation of how these derivations came to be will be explained in the next subsection.

Take note that $P(Y_i \leq j)$ are the cumulative probabilities defined below [33]:

$$P(Y_i \leq j) = 1 - \frac{\exp(\theta_j + \mathbf{x}'_i \boldsymbol{\beta})}{1 + \exp(\theta_j + \mathbf{x}'_i \boldsymbol{\beta})} \quad (2)$$

$$= \frac{1}{1 + \exp(\theta_j + \mathbf{x}'_i \boldsymbol{\beta})} \quad (3)$$

$$\log \left(\frac{P(Y_i \leq j)}{1 - P(Y_i \leq j)} \right) = \theta_j - \mathbf{x}'_i \boldsymbol{\beta} \quad \text{for } j = 0, 1, 2, \dots, J$$

Where

$$\text{Odds} = \frac{P(Y_i \leq j)}{1 - P(Y_i \leq j)}$$

The Odds can be determined by exponentiating the log-odds model.

Although there is only one set of coefficients, there is a different intercept (threshold) parameter for each of the equations. The intercept parameter is dependent on the number of categories the response variable has, therefore, the number of the intercept parameter will be $J - 1$. For example, if my response variable Y has three categories which may be low, medium and high, the model will estimate two intercept parameters and only one set of parameter estimates. Take note the equations estimated will be a set of $J - 1$ binary logit equations. And the way the model is specified, the explanatory variables predict the probability of being in a lower category than in a higher category. Moreover, "In the ordered logit model, there is a continuous, unmeasured latent variable Y^* whose values determine what the observed ordinal

variable Y equals" [32].

6 Statistical Inference

As mentioned above, the parameters of β are estimated by maximizing the log-likelihood [17] function after transforming the depended variable into a logit variable and this optimization can be done in various ways such as the Gradient Ascent, Iterative Reweighted Least Squares (IRLS) and the Newton-Raphson method [18], [19], [20], but we will be looking at the Newton-Raphson method which is a numerical optimization technique.

The Wald test (named after Abraham Wald), will be used to assess the significance of individual predictor variables or sets of predictors in the model. This is a type of test that typically only has 1 degrees of freedom (df) [21]. This test "assesses the constraints of the parameters based on the weighted distance between the unrestricted estimate" [22], [23]. Based on the equation below, the larger the variance of the parameter estimates, the less true the null hypothesis will be.

The Wald test is defined as:

$$W = \frac{\hat{\beta}^2}{\text{SE}(\hat{\beta})^2} \sim \chi^2(1)$$

where $\hat{\beta}$ is the estimated coefficient, and $\text{SE}(\hat{\beta})$ is its standard error.

- $H_0: L\beta = 0$
- $H_a: L\beta \neq 0$

In the above hypothesis testing, L can take on integers ranging from $-\infty$ to ∞

Again, the Hosmer-Lemeshow is a statitsic that can be used to test the the fit of a model. While this test is usually used to assess the goodness of fit for logistic regression models [24], thus it can also be extended to test the multinomial logit models. This test can be achieved by evaluating the fit within each category as stated in the formula below. This is a model named after its creators, which are statisticians David Hosmer and Stanley Lemeshow having $J - 2$ degrees of freedom. This test is typically used in the fields of medical research, biostatistics and many more industries [25]. Although this is a very good test for the purposes of testing the goodness of fit, it also has a limitations which are [26]:

- "Sensitive to the number of groups used"
- "The greater the sample size, the more likely it is to reject the hypothesis if a perfect fit".

The Hosmer-Lemeshow Test is defined as:

$$\chi^2 = \sum_{j=1}^J \frac{(O_j - E_j)^2}{E_j} \sim \chi^2(J - x)$$

where O_j and E_j are the observed and expected frequencies, respectively, in group j , and J is the number of groups and " x " is the number of estimated parameters.

- H_0 : $\beta_1 = \beta_2 = \dots = \beta_p = 0$
- H_a : Not all β_p are equal to 0

These two test statistics have different purposes in the context of evaluating the multinomial logit model. Since the Wald test tests the significance of individual predictor variables, this can help us identify the variables that are not significant, thus we can simply remove these variables and in turn make the model more parsimonious and interpretable. On the other hand, the Hosmer-Lemeshow Test is used to validate the overall fit in the model rather than individual predictor variables. On both of these tests, we are going to test at 1% and 5% level of significance. We might also use the AIC, BIC to determine the goodness of fit of each model, this is just to validate the Hosmer-Lemeshow Test.

But before we do the actual testing, we are firstly going to be looking at other important concepts in statistics. Those concepts are such as the descriptive statistics, which are the fundamental things we have to determine for us to understand the main characteristics of a dataset. Firstly, we are going to create a table for each and every variable in our dataset and see the proportion of the categories within each and every variable. Secondly, a cross tabulation between the response and each explanatory variable is to be created. With this, we can understand the trend of our explanatory variables given the response variable. Since choosing the lowest category as our reference category, this cross tabulation will be comparison in a binary case (e.g 1 vs 0, 2 vs 0 and 3 vs 0). These pairwise comparisons allow us to see how each category in the response variable differs from the reference category.

Since the dataset that we will be analysing is basically or mostly made up of categorical variables, we are going to test if some of those categories within the same explanatory variable are statistically different from each other. Again, if possible that we have an explanatory variable that has a lot of categories which maybe five or more, we are going to perform some statistical analysis to decide if that particular variable should be treated as a qualitative or quantitative variable before we perform our statistical analysis on the data. We can also look at the frequency of the dataset and do a comparative analysis based the response variable and some of the explanatory variables.

7 Application Of The Data In SAS

When it comes to SAS, there are many procedures that use the maximum likelihood estimation that can be used to determine the parameter estimates of the logit models mentioned in this paper. Procedures such as LOGISTIC, SURVEYLOGISTIC, CATMOD and GLIMMIX could be used to achieve the latter. In this paper, we will be using the LOGISTIC procedure because of its simple nature in being implemented and interpreted [27]. The application of this model will be based on the APS dataset (Adolescent Placement Study), which has an impressive number of 508 observations and 11 explanatory variables.

The categorical response variable has three levels of categories ranging from 0-2 which are ordered. Below is the description of the response variable (which is place3). This particular dataset can be used in both the multinomial and ordered logit model. This is because the multinomial logit model does not assume any specific ordered between categories of the response variable while the ordered logit model does assume rank in the response variable.

$$\therefore Y = \begin{cases} 0 & \dots \text{ Outpatient} \\ 1 & \dots \text{ Intermediate Residential Patient} \\ 2 & \dots \text{ Residential Patient} \end{cases}$$

The APS dataset has the following explanatory variables; Age at Admission, Gender, Neuropsychiatric Disturbance, Emotional Disturbance, Danger to Others, Elopment Risk, Length of Hospitalization, Behavioral Symptoms Score, State Custody and History of Violence. For now we are firstly going to examine the all explanatory variables in comparison with the response variable. By using **PROC FREQ** in SAS, we are able to investigate the proportion intermediate residential vs outpatient, and residential vs outpatient.

7.1 DESCRIPTION OF THE DATA

Table 2: Data Description Table

VARIABLE	DESCRIPTION	CODES	NAME	FREQUENCY	PERCENT
Y	Placement Combined	0	PLACE3	259	50.98
-	-	1	-	130	25.59
-	-	2	-	119	23.43
X_1	Age at Admission	12	AGE	93	18.31
-	-	13	-	81	15.94
-	-	14	-	118	23.23

VARIABLE	DESCRIPTION	CODES	NAME	FREQUENCY	PERCENT
-	-	15	-	87	17.13
-	-	16	-	129	25.39
X_2	Race	0 = White	RACE	230	45.28
-	-	1 = Non-white	-	278	54.72
X_3	Gender	0 = Female	GENDER	274	53.94
-	-	1 = Male	-	234	46.06
X_4	Neuropsychiatric Disturbance	0 = None	NEURO	350	68.90
-	-	1 = Mild	-	81	15.94
-	-	2 = Moderate	-	77	15.16
X_5	Emotional Disturbance	0 = Not severe	EMOT	389	76.57
-	-	1 = Severe	-	119	23.43
X_6	Danger to Others	0 = Unlikely	DANGER	54	10.63
-	-	1 = Possible	-	101	19.88
-	-	2 = Probable	-	141	27.76
-	-	3 = Likely	-	212	41.73
X_7	Elopement Risk	0 = No risk	ELOPE	316	62.20
-	-	1 = At risk	-	192	37.80
X_8	Length of Hospitalization	Days	LOSGRP	162	31.89
-	-	1	-	183	36.02
-	-	2	-	163	32.09
X_9	Behavioral Symptoms Score	5	BEHAV	188	37.01
-	-	6	-	106	20.87
-	-	7	-	101	19.88
-	-	8	-	80	15.75
-	-	9	-	33	6.50
X_{10}	State Custody	0 = No	CUSTD	317	62.40
-	-	1 = Yes	-	191	37.60
X_{11}	History of Violence	0 = No	VIOL	121	23.82
-	-	1 = Yes	-	387	76.18

Before getting into analyzing the variables, some of the variables (age, neuro, behav and los) were adjusted for better analysis. Please refer to the appendix to see which variable were adjusted and how they were adjusted. Below is the table that used for primary analysis.

Table 3: Two-Way Frequency Table

PLACE3 VS AGE						
AGE						
PLACE3	12	13	14	15	16	MARGINAL COL
1	22.37 (n = 76)	32.31 (n = 65)	33.33 (n = 96)	44.83 (n = 58)	36.17 (n = 94)	33.42
2	22.37 (n = 76)	26.67 (n = 60)	25.58 (n = 86)	47.54 (n = 61)	36.84 (n = 95)	31.48
PLACE3 VS RACE						
RACE						
PLACE3	0	1				
1	28.49 (n = 179)	37.62 (n = 210)				33.42
2	28.49 (n = 179)	34.17 (n = 199)				31.48
PLACE3 VS GENDER						
GENDER						
PLACE3	0	1				
1	30.88 (n = 217)	36.62 (n = 172)				33.42
2	27.54 (n = 207)	36.26 (n = 171)				31.48
PLACE3 VS NEURO						
NEURO						
PLACE3	0	1	2			
1	31.48 (n = 270)	40.32 (n = 62)	35.09 (n = 67)			33.42
2	30.19 (n = 265)	33.93 (n = 56)	35.09 (n = 57)			31.48
PLACE3 VS EMOT						
EMOT						
PLACE3	0	1				
1	30.13	44.83				33.42

	(n = 302)	(n = 87)				
2	29.19 (n = 298)	40 (n = 80)				31.48
PLACE3 VS DANGER						
DANGER						
PLACE3	0	1	2	3		
1	14.29 (n = 49)	41.03 (n = 78)	43.64 (n = 110)	28.29 (n = 152)		33.42
2	10.64 (n = 47)	33.33 (n = 69)	33.33 (n = 93)	35.50 (n = 169)		31.48
PLACE3 VS ELOPE						
ELOPE						
PLACE3	0	1				
1	31.25 (n = 256)	37.59 (n = 133)				33.42
2	25.42 (n = 236)	41.55 (n = 142)				31.48
PLACE3 VS LOSGRP						
LOSGRP						
PLACE3	0	1	2			
1	28.57 (n = 154)	34.39 (n = 157)	41.03 (n = 78)			33.42
2	6.78 (n = 118)	20.16 (n = 129)	64.89 (n = 131)			31.48
PLACE3 VS BEHAV						
BEHAV						
PLACE3	5	6	7	8	9	
1	26.32 (n = 171)	39.77 (n = 88)	31.75 (n = 63)	48.00 (n = 50)	35.29 (n = 17)	33.42
2	11.89 (n = 143)	25.35 (n = 17)	46.91 (n = 81)	53.57 (n = 56)	59.26 (n = 27)	31.48
PLACE3 VS CUSTD						
CUSTD						

PLACE3	0	1				
1	8.59 (n = 256)	81.20 (n = 133)				33.42
2	20.68 (n = 295)	69.88 (n = 83)				31.48
PLACE3 VS VIOL						
VIOL						
PLACE3	0	1				
1	24.53 (n = 106)	36.75 (n = 283)				33.42
2	15.79 (n = 95)	36.75 (n = 283)				31.48

From investigating the column percentages using "**PROC FREQ**" output, we can see that age has on average a positive trend, moreover, we can notice that the likelihood of being a residential patient increases as age increases as well. Now as for Race, we can see that people that are not white tend to be residential patients, with 22% of white individuals being residential patients compared to the 24% of non-white. In the case of gender, males are more likely to be residential patients compared to females with the proportions 21% to 27%. With the neuro variable and looking at the marginal proportions, there appears to be a negative trend over the categories with a sharp downfall from category 0 to 1. When it comes to the variable danger, when analyzing the comparison categories between 0 and 2, we get for every unit increase in danger, the odds of being an outpatient and residential patient both increase linearly. With the Length of Hospitalization that has been grouped into three categories, there seems to be an exponential increase, meaning with residential patients tend to stay longer at a hospital. Judging behavioral symptoms score at face value, there is on average a positive trend in this variable. The variables that were not interpreted here basically are simple to interpret.

7.2 BINARY LOGIT MODEL RESULTS

Since our response variable has three categories, it will be examined how the odds of being a residential or intermediate residential patient versus being an outpatient is influenced by the explanatory variables. In this section, this will be achieved by fitting two logistic regression models and is discussed below.

By using **PROC LOGISTIC** in SAS where you want to compare two categories, you use the following code: Where $\neq$ means **not equal to**, and the value "**x**" can take on categories 0, 1 or 2 of the response variable.

```

1 proc logistic data = new_aps;
2 where place3 ^= x;
3 model place3 (REF = '0') = age race gender neuro emot danger elope
   losgrp behav custd viol / aggregate scale = none;
4 run;

```

As we go forward, keep in mind that the reason why the estimates have only one degree of freedom is because in the **PROC LOGISTIC** procedure we fitted a slope on all the explanatory variables. When looking at the **ANALYSIS OF MAXIMUM LIKELIHOOD ESTIMATES**, this procedure will only estimate $K - 1$ equations, meaning if a response variable has two categories, only 2-1 equations will be estimated, and below are the results of the estimates.

Now since our reference category in this paper is 0, the first set of our results is basically a comparison equation between category 1 and 0, and the second set a comparison equation of categories 2 and 0. The results of are as follows:

Parameter	PLACE3	DF	Estimate	Standard Error	Chi-Square	Wald	Pr > ChiSq
Intercept	1	1	-7.9609	2.0488	15.0978	0.0001	
AGE	1	1	0.2157	0.1187	3.3060	0.0690	
RACE	1	1	0.3791	0.3311	1.3109	0.2522	
GENDER	1	1	0.1348	0.3712	0.1319	0.7165	
NEURO	1	1	0.4456	0.2259	3.8912	0.0485	
EMOT	1	1	0.1857	0.4005	0.2149	0.6429	
DANGER	1	1	0.1245	0.2170	0.3292	0.5662	
ELOPE	1	1	-0.2550	0.3699	0.4752	0.4906	
losgrp	1	1	0.3636	0.2224	2.6728	0.1021	
BEHAV	1	1	0.2014	0.1617	1.5520	0.2128	
CUSTD	1	1	4.1424	0.3723	123.8172	<.0001	
VIOL	1	1	0.2272	0.4854	0.2190	0.6398	

Figure 1: Maximum Likelihood Estimates for 1 vs 0

Parameter	PLACE3	DF	Estimate	Standard Error	Chi-Square	Wald	Pr > ChiSq
Intercept	2	1	-11.5634	2.0455	31.9589	<.0001	
AGE	2	1	0.2552	0.1129	5.1065	0.0238	
RACE	2	1	0.1343	0.3109	0.1866	0.6658	
GENDER	2	1	0.4839	0.3530	1.8784	0.1705	
NEURO	2	1	0.1402	0.2061	0.4625	0.4965	
EMOT	2	1	0.2123	0.3890	0.2977	0.5853	
DANGER	2	1	-0.2952	0.2021	2.1330	0.1442	
ELOPE	2	1	0.5166	0.3253	2.5224	0.1122	
losgrp	2	1	1.6096	0.2292	49.2989	<.0001	
BEHAV	2	1	0.6455	0.1530	17.7860	<.0001	
CUSTD	2	1	2.1944	0.3907	31.5465	<.0001	
VIOL	2	1	0.5007	0.4950	1.0230	0.3118	

Figure 2: Maximum Likelihood Estimates for 2 vs 0

Looking at the comparison between 1 and 0, we see that only neuro and state custody are the significant parameter estimates in this case, leaving the rest insignificant. And in the results between comparison 2 and 0, we get that age, losgrp, behav and custody are significant. With that being said, it will be interesting to investigate the significant variables as go further. All these significant variables were tested at 5% level of significance.

Looking at all the parameter estimates, besides the intercept and elope, all the other explanatory variables have a positive trend. Now looking at the comparison between 2 and 0 for age, it basically means that there is a positive trend between age and the likelihood of being a residential patient. Moreover, looking at the comparison between 1 vs 0 for neuro, since its coefficient is positive, individuals with a neurological condition are more likely to be classified under intermediate residential patient. So trend basically implies a constant pattern that emerges of which is often used to describe the relationship between variables.

From the results we have gathered so far, we might think the results are not reliable, for this we can actually use two goodness of fit statistics namely the **DEVIANCE** and the **PEARSON CHI-SQUARE**. These two test statistics actually measure whether the observed data matches the expected distribution of model. The results from these two goodness of fit statistics are as follows:

Deviance and Pearson Goodness-of-Fit Statistics				
Criterion	Value	DF	Value/DF	Pr > ChiSq
Deviance	254.5959	364	0.6994	1.0000
Pearson	364.0231	364	1.0001	0.4898
Number of unique profiles: 376				

Figure 3: Goodness Of Fit
Statistic 1 vs 0

Deviance and Pearson Goodness-of-Fit Statistics				
Criterion	Value	DF	Value/DF	Pr > ChiSq
Deviance	259.9346	349	0.7448	0.9999
Pearson	348.8162	349	0.9995	0.4927
Number of unique profiles: 361				

Figure 4: Goodness Of Fit
Statistic 2 vs 0

In summary, when you get a low p-value from these statistics, it means there is a poor fit, thus reject the model. A high p-value, good fit, retain the model. Looking solely at the deviance statistic, we see that the p-value of the deviance in the first and second results is very high, meaning that those two binary models there is an adequate fit, but this is all dependant on the level of significance you are testing at. Moreover, the degrees of freedom for both test statistic are calculated by taking the number of unique profiles and subtracting that by the number of estimated parameters by that model, and the number of unique profiles represent distinct patterns of the predictor variables in the dataset. In addition, this statistic might not be reliable and this is because when a logit model has categorical predictor variables, that have many levels e.g each categorical variable has 10 levels, their interaction will result in 100 cells and if the sample size is not enough to populate these cells, the chi-square test will suffer to assess the model fit.

The odds ratio output below is for the the comparison between categories 2 and 0, these estimated odds ratio were calculated by exponentiating the parameter estimates ($\exp(\hat{\beta})$) estimated using the Maximum Likelihood Estimation technique. A few variables will be used as an example of how these odds ratios can be interpreted. The reason for choosing this comparison output is because we would like to compare the lowest category to the highest, as well as this output will make it easier to compare this type of output in this model with other models.

Odds Ratio Estimates				
Effect	PLACE3	Point Estimate	95% Wald Confidence Limits	
AGE	2	1.291	1.034	1.611
RACE	2	1.144	0.622	2.104
GENDER	2	1.622	0.812	3.241
NEURO	2	1.150	0.768	1.723
EMOT	2	1.236	0.577	2.650
DANGER	2	0.744	0.501	1.106
ELOPE	2	1.676	0.886	3.171
losgrp	2	5.001	3.191	7.837
BEHAV	2	1.907	1.413	2.574
CUSTD	2	8.974	4.173	19.300
VIOL	2	1.650	0.625	4.353

Figure 5: Odds Ratio Estimates For 2 vs 0

Interpreting the odds ratio for the variable age; for every increase in one year, the odds to be a residential patient increases by 29%. With regards to the race variable, individuals from the Non-White category are 14% more likely to be placed in as residential patients than outpatients. So now the odds of most of these variables can generally be explained in the same way, but there arises a significant problem, even visible with a naked eye with regards to the **CUSTD**, **LOSGRP** variables having the odds ratio of 8.974 and 5 respectively. This now tells us that significant variables most probably have a tendency to have higher odds ratios than usual. This ratio in summary tells that these variables have a very strong relationship with the response variable, and to explain these odds, these odds ratio basically mean that individuals that are state custody have 9 times higher odds of being residential patients than being outpatients, this interpretation is the same for the LOSGRP variable.

Now referring back to our theoretical concepts, we did mention before that when estimates are on a log scale like the estimates we have in our results, exponentiating them can also give us the estimated odds ratio of that particular variable in question. This can be achieved by using the ESTIMATE statement in SAS: To test this, we are going to use the comparison between categories 2 and 0.

```
ESTIMATE "AGE" age 1 / exp;
```

From this line of code, you get the following results:

Estimate					
Label	Estimate	Standard Error	z Value	Pr > z	Exponentiated
AGE	0.2552	0.1129	2.26	0.0238	1.2908

Figure 6: Odds Ratio Estimate For Age 2 vs 0

Now what the above statement does is this; $e^{0.2552} = 1.29$.

Results such as these can be interpreted the same way as interpreted above. Again, from all of this

we can also calculate the probabilities given certain conditions.

And now just to add some context in determining probabilities, what we are going to do now is create a mini model taking only a few variables from our dataset using categories 2 and 0 of our response variable to explain the context behind calculating such probabilities. The selected variables are age, race and gender. Using the **proc logistic** procedure, and by making use of **ilink** with the **exp** option, we can add the following code after the model statement:

```
1 proc logistic data = new_aps;
2 where place3 ^=1;
3 model place3 (REF = '0') = age race gender / aggregate scale = none;
4 ESTIMATE "PROB" intercept 1 age 12 race 1 gender 0 / exp ilink;
5 run;
```

Where this code is basically estimating the log-scale estimate, odds and probability (mean) of an individual who is 12 years of age, and a non-white female.

Estimate							
Label	Estimate	Standard Error	z Value	Pr > z	Mean	Standard Error of Mean	Exponentiated
PROB	-1.4641	0.2707	-5.41	<.0001	0.1878	0.04131	0.2313

Figure 7: Probability Estimation Example

From this output, we can see that there is an 18.78% chance that a 12 year old, non-white female will be placed as a residential patient. This can also be interpreted in terms of odds, where the odds of 0.23 suggest that this individual is less likely to be a residential patient than an outpatient. Keep in mind the same thought process will be used in the multinomial logit model and compare if the results are almost similar. But in the multinomial logit model we will be using **SAS proc iml** since it is not possible to properly use the exp and the ilink functions in the multinomial logit case.

7.3 MULTINOMIAL LOGIT MODEL RESULTS

As it has been mentioned before that the **MULTINOMIAL LOGIT MODEL** is basically a build up of the **BINARY LOGIT MODEL**. In this section, instead of fitting two separate logit models, we will now analyse the categories of the response variable by fitting a multinomial logit model.

EFFECT	1 vs 0	2 VS 0	2 VS 1
INTERCEPT	-8.0199**	-11.9625**	-3.9426*
AGE	0.2392*	0.2642*	0.0250
RACE	0.4173	0.2167	-0.2006
GENDER	0.4159	0.3842	-0.0318
NEURO	0.4831*	0.2329	-0.2502
EMOT	0.1908	0.1746	-0.0161
DANGER	0.1057	-0.2124	-0.3182
ELOPE	-0.1078	0.5522	0.6600*
LOSGRP	0.3584	1.7329**	1.3744**
BEHAV	0.1428	0.6351**	0.4922**
CUSTD	4.1504**	2.3793**	-1.7711**
VIOL	0.1054	0.3828	0.2775

Table 4: Maximum Likelihood Estimates Table

From the results shown in the table above, the estimates that have double asterisks are significant at 1% level of significance, while the one's with a single asterisk are significant at 5% level of significance. Moreover, looking at the results at face value, we can say that they are almost similar to the one's in the binary logit model. From the output above, the estimates in the 3rd column of category 2 vs 1 were obtained by re-running the **proc logistic** procedure and making "1" the reference category. And since that is the case, it would also be good to show the mathematical background of that column as well:

$$\begin{aligned}\log\left(\frac{P(Y=2)}{P(Y=1)}\right) &= \log\left(\frac{\frac{P(Y=2)}{P(Y=0)}}{\frac{P(Y=1)}{P(Y=0)}}\right) \\ &= \log\left(\frac{P(Y=2)}{P(Y=0)}\right) - \log\left(\frac{P(Y=1)}{P(Y=0)}\right)\end{aligned}$$

Now from this, we can actually obtain the same results without re-running the model.

We can also use the test statement in the **proc logistic** procedure after the **"model"** statement to test if the coefficients in the first column are different to the corresponding coefficients in the second column. The variable age will be used as an illustration:

```
1 AGE: TEST age_1 = age_2;
```

Now take note that when we append with `_1`, this basically represents the variable age in the first equation (column), testing if it is equal to age in the second equation (column) which is appended by `_2`.

Linear Hypotheses Testing Results			
Label	Wald Chi-Square	DF	Pr > ChiSq
AGE	0.0512	1	0.8211

Figure 8: Significance Of Age Variables

From this output and testing at 5% level of significance, we get that the two coefficients at hand

actually do differ significantly from each other. This can also be done for all the other explanatory variables.

Now these odds ratio output for the multinomial case were obtained by using the code below:

```
1 ODDSRATIO age;
```

In which the code gives us the odds ratio for both comparison categories of 1 vs 0 and 2 vs 0.

Odds Ratio Estimates and Wald Confidence Intervals			
Odds Ratio	Estimate	95% Confidence Limits	
PLACE3 1: AGE	1.270	1.028	1.569
PLACE3 2: AGE	1.302	1.055	1.608

Figure 9: Two Odds Ratio Estimates

Also, notice that on the picture above, the odds ratio of age where place3 = 2 (This is basically the output between comparison categories 2 vs 0) is 1.302 for this model, where as the odds ratio in the previous model we was 1.2908.

Now the graph below is effect plot of the predicted probabilities that were determined using the following codes in SAS after the model statement and before the run statement:

```
1 output out = predicted predprobs = I;
2 effectplot fit (x = age) / noobs nolimits;
```

Now the first line of code determines the predicted probabilities for each category of the response variable and stores them in a dataset called "predicted". Now the second line of code creates a plot showing fitted probabilities of the response variable as a function of the "age" variable.

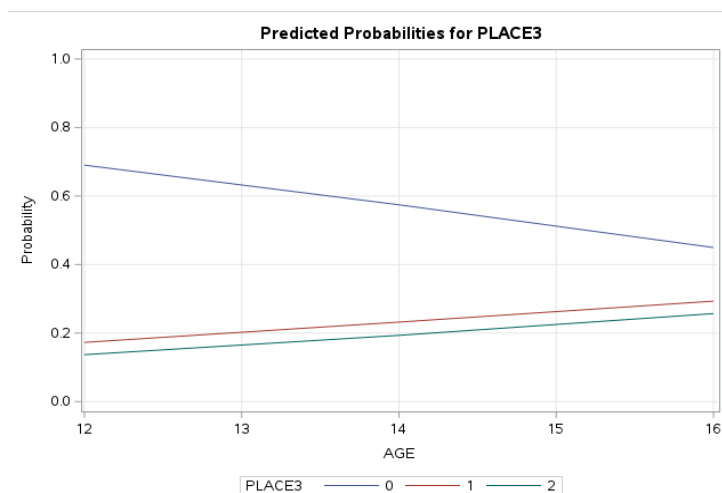


Figure 10: Predicted Probabilities Trend

Looking at the plot above with the red and green lines, the probability to be an intermediate residential or residential patient increases as a patient gets older.

As stated before that a mini model using the multinomial logit model with variables age, race and gender will be created to compare the results in the binary logit model to this model. The results to be used are the comparison between 2 and 0. The following lines of code were used:

```

1 proc logistic data = new_aps outest = siya;
2 model place3 (REF = '0') = age race gender / LINK = GLOGIT aggregate
   scale = none;
3 run;
4
5 proc iml;
6 use siya;
7 read all var{Intercept_2, AGE_2, RACE_2, GENDER_2} into data_2; print
   data_2;
8 Intercept_2 = data_2[1,1]; print Intercept_2;
9 AGE_2 = data_2[1,2]; print AGE_2;
10 RACE_2 = data_2[1,3]; print RACE_2;
11 GENDER_2 = data_2[1,4]; print GENDER_2;
12 log_scale = Intercept_2 * 1 + AGE_2 * 12 + RACE_2 * 1 + GENDER_2 *
   0; print log_scale;
13 odds = exp(log_scale); print odds;
14 P = odds / (1 + odds); print p;

```

From this code, we get results almost similar to the ones we got in the binary logit model. We firstly determined the parameter estimate which was -1.437851. Secondly, we determine the odds ratio which was 0.2374374, using the theoretical formula for odds calculation stated in the binary logit model section, we got a probability of 0.1918783. The interpretation is similar to the calculations done in the binary model section but what is important, we can see that this further solidifies the fact the multinomial logit model is a build up of binary logit models. Keep in mind that the reason why we multiply our estimates with categories 12, 1 and 0, is because we want to estimate the probability of an individual who is 12 years of age, who is a Non-White female.

7.4 ORDERED LOGIT MODEL RESULTS

When it comes to this model, instead of jumping to determine our parameter estimates, we can first try to determine if the proportional odds assumption is applicable to our model. The proportional odds assumption implies that the effects of being an intermediate residential or residential patient are the same across different thresholds of the ordinal outcome variable. To do this, we can go back to the multinomial

logit model and use the test statement to see if the estimates appended by _1 and _2 are equal in that particular explanatory variable. This can be achieved in SAS by using the test statement after the model statement, take note of the code below:

```
1 Proportional_Odds_Test: test age_1=age_2, race_1=race_2, gender_1=
    gender_2, neuro_1=neuro_2, emot_1=emot_2,
2 danger_1=danger_2, elope_1=elope_2, losgrp_1=losgrp_2, behav_1=behav_2,
    custd_1=custd_2, viol_1=viol_2;
```

And from the code above, we get the following output:

Linear Hypotheses Testing Results			
Label	Wald Chi-Square	DF	Pr > ChiSq
Proportional_Odds_Test	73.4118	11	<.0001

Figure 11: Proportional Odds Test

If the p-value is less than 0.05, then the assumption is violated and in this case we have to consider other alternative models that will suite our dataset well. And from this we can see that the assumption is violated.

When you implement the "**PROC LOGISTIC**" for the ordinal model, we automatically get the **Score Test for the Proportional Odds Assumption** result, although not a reliable estimate, it can be used to further prove that the assumption is violated.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
171.5940	11	<.0001

Figure 12: Score Test

The test statistic is significant, which suggests that the proportional odds assumption of the predictors in the modeled relationship cannot be assumed. In most cases when using the "score test for the proportional odds assumption", the null hypothesis is often rejected because of large datasets with a lot of explanatory variables which is why it might not be a reliable estimate. When this is the case, it is sometimes useful to assess this proportional odds assumption visually for a given explanatory variable. To achieve this, we are going to use the "**MOSAIC PLOT**", if this plot shows "the proportion of observations in each level of the response is not consistent across each level of the explanatory variable", then it means that the proportional odds assumption is violated [35]. The variable "**CUSTD**" is used as an illustrative example.

```
1 proc freq data=new_aps;
```

```

2 table place3*custd / plots = mosaicplot;
3 run;

```

The figure looks like this and we can clearly see that the assumption is violated on this variable.

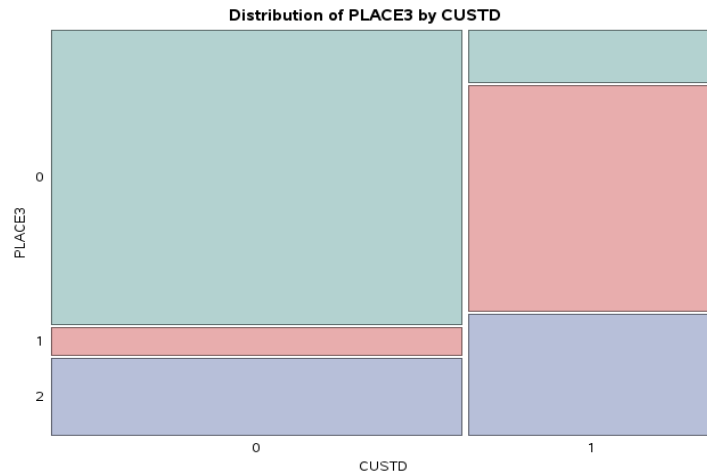


Figure 13: Mosaic Plot

After further analysis on all of the other explanatory variables against the response variable, it was found that the variables "loggrp", "behav" and "custd" were the one's that showed that the proportion of observations in each level of the response are not consistent across each level of the explanatory variable. And again, recall that in the multinomial logit model, these variables were the ones that had very high odds ratios, if this is now the case, it can also be said to make sure that the proportional odds assumption applies in this model, it would be best to first look at the odds ratios of the variables you might be investigating to see if they are not extremely high. Thus it was decided to exclude them from the model to see if the proportional odds assumption would apply if removed.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
7.0726	8	0.5288

Figure 14: Score Test After Removal

In conclusion, we see that with the problematic variables removed (variables with very high odds ratios), the proportional odds assumption is not violated when the variables are removed.

Since now we have determined the parsimonious model, the following code was used to determine the Maximum Likelihood Estimates:

```

1 proc logistic data = new_aps;
2 model place3 (desc) = age race gender neuro emot danger elope viol /
    aggregate scale = none;

```

```
3 run ;
```

Notice that the function **"DESC"** was used in the model statement. One thing that had to be understood about this model is that the probabilities modeled are cumulated over the lower ordered values, now the above stated function is used so that we can compare the odds ratios of the three models in this paper. Now instead of the probabilities being modeled over lower ordered values, the probabilities are going to modeled over the highest category of the response variable. Now in this case, we will model the odds of being a residential patient. Now below are the Maximum Likelihood Estimates.

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	2	1	-8.0305	1.0003	36.3416	<.0001
Intercept	1	1	-4.7851	0.9867	23.5189	<.0001
AGE		1	0.2289	0.0636	12.9540	0.0003
RACE		1	0.3363	0.1758	3.6588	0.0558
GENDER		1	0.3453	0.1892	3.3283	0.0681
NEURO		1	0.0748	0.1174	0.4055	0.5243
EMOT		1	0.4941	0.2075	5.6718	0.0172
DANGER		1	0.0534	0.1061	0.2535	0.6146
ELOPE		1	0.6344	0.1824	12.0928	0.0005
VIOL		1	0.8587	0.2609	10.8315	0.0010

Figure 15: Maximum Likelihood Estimation

From this output, it shows that age, emot, elope and viol are all significant at 5% level of significance. Now notice that this parsimonious model has trends similar to that of the multinomial logit model. Just as when you have to explain the trend of age, you can say that since the model reflects a positive relationship between age and the likelihood of being in a higher category.

Now just like in the binary logit model, these parameter estimates are given on a log-scale, thus when exponentiating the variable in question using the following line of code below the model statement we will get the odds ratio of the variable in question:

```
1 ESTIMATE "AGE" age 1 / exp ;
```

The following is now the odds ratios of the explanatory variables:

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
AGE	1.257	1.110	1.424
RACE	1.400	0.992	1.975
GENDER	1.412	0.975	2.047
NEURO	1.078	0.856	1.357
EMOT	1.639	1.091	2.461
DANGER	1.055	0.857	1.299
ELOPE	1.886	1.319	2.697
VIOL	2.360	1.415	3.936

Figure 16: Odds Ratio Estimates

In this model, we have to keep in mind that this model assumes that the odds ratios are constant across different response levels, hence that is why we are getting one odds ratio for each explanatory variable as compared to the multinomial logit model where we had two odds ratios for each explanatory variable. Now to interpret the variable age; for every increase in one year, the odds of being residential patient increases by 26%. And now considering the variable race with an odds ratio of 1.4, Non-White individuals are 40% more likely than White individuals to become residential patients.

Now using this very same code in the ordinal model case, we would like to see if as age increases, the probability of being an intermediate residential and residential patient also increases just as we saw in the multinomial logi model.

```

1 output out = predicted predprobs = I;
2 effectplot fit (x = age) / noobs nolimits;

```

From the above code, this is the output we get:

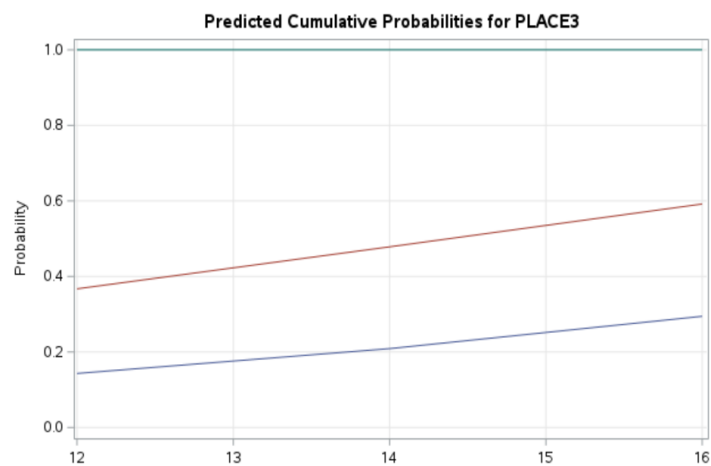


Figure 17: Predicted Probabilities Ordinal case

While looking at the graph, note that the blue line is place3 = 2, the red is place3 = 1 and the green is place3 = 0. From this graphical output, we can arrive to the same conclusion that the probability to

be an intermediate residential or residential patient increases as a patient gets older.

8 CONCLUSION

Although not the overall purpose of this paper, there emerged a very insightful link between the two models in consideration where in the Multinomial Logit Model, variables that had very high odds ratios directly impacted the proportional odds assumption in the Ordinal Logit Model. Now this in turn led us to find a parsimonious and more interpretable model.

In doing this research, the major key distinction here lies in the very nature of our dataset (Ordered or Unordered Response Variable) and the assumptions that underlying each model. Now summarising all the important results, in the Multinomial Logit Model, there were a few parameter estimates that had a negative trend while our parsimonious Ordinal Logit Model only had parameter estimates that had a positive trend. Moreover, the odds ratios of almost each and every explanatory variable in the multinomial logit model was close to those that are in the ordinal logit model, thus leading to the same conclusions about the data. Moreover, this was further proved by using the predicted probabilities as a function of age to see if both in the multinomial logit case and in the ordinal model case will lead to the same graphical conclusions of which it did.

But there arises a problem with the ordinal logit model, as we know that if some of the variables that were removed were not removed, the Ordinal Logit Model was not going to apply in this dataset, thus making the multinomial logit model the best model to model this type of dataset even though the dataset had inherent ordering in it. But on the other hand, even though the dataset had a response variable with more than three categories with inherent ordering, it would be the most easily interpretable model compared to the multinomial logit model where interpretation becomes harder in this case.

Based on the finding in this paper, using the Multinomial Logit Model is advised when analysing this type of dataset. This is because the model offers a more flexible approach to modeling complex relationships between the response and the explanatory variables. Moreover, the model captures the effect of all the explanatory variables.

This paper also contributes to the existing literature on modeling categorical data, thus enhancing the importance of model selection based on your data and research questions at hand.

With all that has been said, it is recommended that we do further investigations on what causes high odds ratios on variables, and what impacts does it have on models and what can be done about. And moreover, future research should also explore the interactions of significance parameters to better understand how significant is their relationship.

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Appendix

Below is the code used to adjust some of the explanatory variable for better analysis.

```
1 data new_aps;  
2 set aps;  
3 age = round(age);  
4 if age = 11 then age = 12;  
5 if age = 18 then age = 17;  
6 if age = 17 then age = 16;  
7 if neuro = 3 then neuro = 2;  
8 if 0 <= behav <= 5 then behav = 5 ;  
9 if 0 <= los < 7 then losgrp = 0;  
10 else if 7 <= los < 14 then losgrp = 1;  
11 else if los >= 14 then losgrp = 2;  
12 run;
```

Below is the google drive link for the complete code done for this research topic;

Research SAS Code.